

HW3

October 16, 2025

1 HW3 — Make the 2D Photo Pop Out

Total Points: 100

Due: Thu, Oct 16 @ 11:59 PM ET

Goal: Backproject a single RGB image into a colored 3D point cloud.

In this assignment, you will: 1. Build a camera intrinsics matrix from image size and FOV. 2. Convert relative depth (Depth-Anything v2) to metric depth. 3. Backproject pixels to camera coordinates. 4. Color each 3D point from the RGB image. 5. Export a point cloud.

What to submit: - The original notebook (HW3.ipynb) - A PDF version of the notebook (HW3.pdf) - The exported point cloud (emory_campus_pcd.ply) - A screenshot of the point cloud opened in MeshLab (emory_campus_pcd_visualized.png) - The point cloud built from your favorite outdoor scene photo (favorite_scene_pcd.ply) - A screenshot of the point cloud built from your photo (favorite_scene_pcd_visualized.png)

```
[ ]: import os, math
import numpy as np
import cv2
import matplotlib.pyplot as plt
import imageio
```

```
[ ]: plt.rcParams["figure.figsize"] = (5,5)

def show(img, title=None, gray=False):
    if gray or (img.ndim==2):
        plt.imshow(img, cmap='gray')
    else:
        if img.ndim==3 and img.shape[2]==3:
            img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
        plt.imshow(img)
    if title: plt.title(title)
    plt.axis('off'); plt.show()
```

1.1 Part A. Build the intrinsic matrix (20 points)

1.1.1 Explanation

We'll use the pinhole camera model with square pixels and the principal point at the image center. Since we don't know the focal lengths (f_x) and (f_y) a priori, we will derive them from the field of view (FOV) and the image size.

Intrinsic matrix

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \quad c_x = \frac{W}{2}, \quad c_y = \frac{H}{2}$$

- f_x, f_y : focal lengths in pixels along the x and y image axes (how many pixels correspond to one unit of distance at unit depth). Assuming square pixels, $f_y = f_x$.
- c_x, c_y : the principal point in pixel coordinates (where the optical axis intersects the sensor). In this homework, we will assume the principal point at the image center.

What is FOV?

The **field of view (FOV)** is the angular extent of the scene the camera sees.

1.1.2 Q1-1. Derive f_x from horizontal FOV (FOV_x).

Hint: Use triangle. What is $\tan(\text{FOV}_x / 2)$?

Write Your Answer Here:

$$f_x = \frac{c_x}{\tan(\frac{\text{FOV}_x}{2})}$$

###Q1-2. Implement `build_intrinsics(W, H, fov_deg)` Write a function that returns the pinhole intrinsics matrix K given the image shape and FoV.

```
[3]: def build_intrinsics(W:int, H:int, fov_deg:float=None) -> np.ndarray:

    if fov_deg is None: fov_deg = 60.0
    K = None

    ### YOUR CODE HERE
    cx = W / 2
    cy = H / 2

    fov_rad = math.radians(fov_deg)
    fx = cx / math.tan(fov_rad / 2)
    fy = fx

    K = np.array([[fx, 0, cx],
                  [0, fy, cy],
                  [0, 0, 1]], dtype=np.float64)
    ### END YOUR CODE

    return K
```

```
[4]: def check(K, K_expected):
      np.testing.assert_allclose(K, K_expected, rtol=0, atol=1e-9)
      print("Passed!")
```

```
[5]: W, H, FOV = 200, 100, 90
      K = build_intrinsics(W, H, FOV)
      K_expected = np.array([[100.0, 0.0, 100.0],
                             [ 0.0, 100.0, 50.0],
                             [ 0.0, 0.0, 1.0]], dtype=np.float64)
      check(K, K_expected)
```

Passed!

1.2 Part B. Compute Metric Depth from Relative Depth (30 points)

1.2.1 Explanation

Depth-Anything v2 predicts **relative depth** that behaves like **inverse depth** (also called disparity): closer points should have larger “depth” responses than farther points. Many monocular/stereo methods model distance this way because: - In pinhole geometry, stereo **disparity** $1 / Z$ (Z = metric depth). - Inverse depth compresses very large distances and is numerically easier to learn.

We convert it to **metric depth** by assuming the model’s output is an **affine transform of inverse depth**:

- **Model:** $Z = 1 / (A + B * y)$
 - y = normalized relative depth in $[0, 1]$.
 - Z = metric depth (meters).
 - A, B = unknown calibration scalars.
- **Percentile calibration:**
 - 1) Pick two **percentiles** of y (e.g., $p_{lo}=5\%$ for “far-ish background”, $p_{hi}=95\%$ for “near-ish foreground”).
 - 2) Assign **plausible metric depths** to those two percentiles (e.g., $ZL_target=2.5$ m, $ZH_target=0.8$ m).
 - 3) Solve the two linear equations in inverse depth:

$$1/ZL_target = A + B * yL$$

$$1/ZH_target = A + B * yH$$
 which gives A and B .
 - 4) Compute metric depth per pixel:

$$depth_m = 1 / (A + B * y)$$

Read the relative (inverse) depth map predicted by [Depth Anything V2](#).

```
[9]: # Input file path
      rgb_path = "emory-campus.png"
      depth_raw = "emory-campus-depth-raw.png"

      # --- Load RGB input ---
```

```

rgb_bgr = cv2.imread(rgb_path, cv2.IMREAD_COLOR)
assert rgb_bgr is not None, "Could not read RGB"
Hc, Wc = rgb_bgr.shape[:2]
show(rgb_bgr, "RGB")

# --- Load depth as relative (0..1) ---
depth_rel = None

r16 = cv2.imread(depth_raw, cv2.IMREAD_UNCHANGED)
if r16 is not None:
    if r16.dtype != np.uint16:
        raise ValueError("16-bit PNG must be uint16.")
    nonzero = r16[r16>0]
    denom = float(nonzero.max()) if nonzero.size else 1.0
    depth_rel = r16.astype(np.float32)/max(denom,1.0)
    show((np.clip(depth_rel,0,1)*255).astype(np.uint8), "Relative depth",
    ↪gray=True)

Hd, Wd = depth_rel.shape[:2]

# Align RGB to depth resolution
if (Hd,Wd)!=(Hc,Wc):
    rgb_bgr = cv2.resize(rgb_bgr, (Wd,Hd), interpolation=cv2.INTER_LINEAR)
    Hc, Wc = rgb_bgr.shape[:2]

```

RGB



Relative depth



```
[10]: def normalize_depth(x):
        x = x.astype(np.float32)
        mn, mx = float(x.min()), float(x.max())
        return (x - mn) / (mx - mn + 1e-8)
```

###Q2-1. Convert inverse depth to metric depth.

Compute the metric depth.

```
[11]: # --- Depth Anything v2 calibration: TrueDepth = 1 / (A + B * y), y in [0,1]
        ↳ ---
        y = normalize_depth(depth_rel) # norm_prediction

        # Pick two percentiles and assign plausible metric depths (m)
        p_lo, p_hi = 5.0, 95.0 # far-ish, near-ish
        yL, yH = np.percentile(y, [p_lo, p_hi])
        ZL_target = 2.5 # meters (far background)
        ZH_target = 0.8 # meters (near foreground)

        invZL, invZH = 1.0/ZL_target, 1.0/ZH_target

        ### YOUR CODE HERE
        B = (invZH - invZL) / (yH - yL)
        A = invZH - B * yH

        depth_m = 1.0 / (A + B * y)
        ### END YOUR CODE
```

```
[12]: # Cleanup
valid = np.isfinite(depth_m) & (depth_m > 1e-6)
if valid.any():
    lo, hi = np.percentile(depth_m[valid], [1.0, 99.0])
    valid &= (depth_m >= lo) & (depth_m <= hi)

# Remove near-white background
rgb = cv2.cvtColor(rgb_bgr, cv2.COLOR_BGR2RGB)
valid &= (np.mean(rgb, axis=2) < 245)

print(f"Depth A={A:.6f}, B={B:.6f}")
print("Valid pixels:", int(valid.sum()), "/", valid.size)

# Range-normalized visualization
rng = np.ptp(depth_m[valid]) if valid.any() else 1.0
viz = (np.clip((depth_m - depth_m[valid].min()) / (rng + 1e-8), 0, 1) * 255).
      .astype(np.uint8)
show(viz, "Metric depth", gray=True)
```

Depth A=0.400000, B=1.932282
Valid pixels: 1462743 / 1500000

Metric depth



1.3 Part C. 2D→3D Backprojection (30 points)

1.3.1 Explanation

- 1) From image coordinates u, v , first map back to camera coordinate system by shifting the origin.
- 2) Then move from the image plane at depth f to the normalized image plane at depth 1.

3) Finally, scale along the ray by the metric depth to recover the 3D point.

###Q3-1. Convert inverse depth to metric depth.

```
[13]: FOV_DEG = 120.0
K = build_intrinsics(Wd, Hd, fov_deg=FOV_DEG)
fx, fy, cx, cy = K[0,0], K[1,1], K[0,2], K[1,2]
print("K =\n", K)

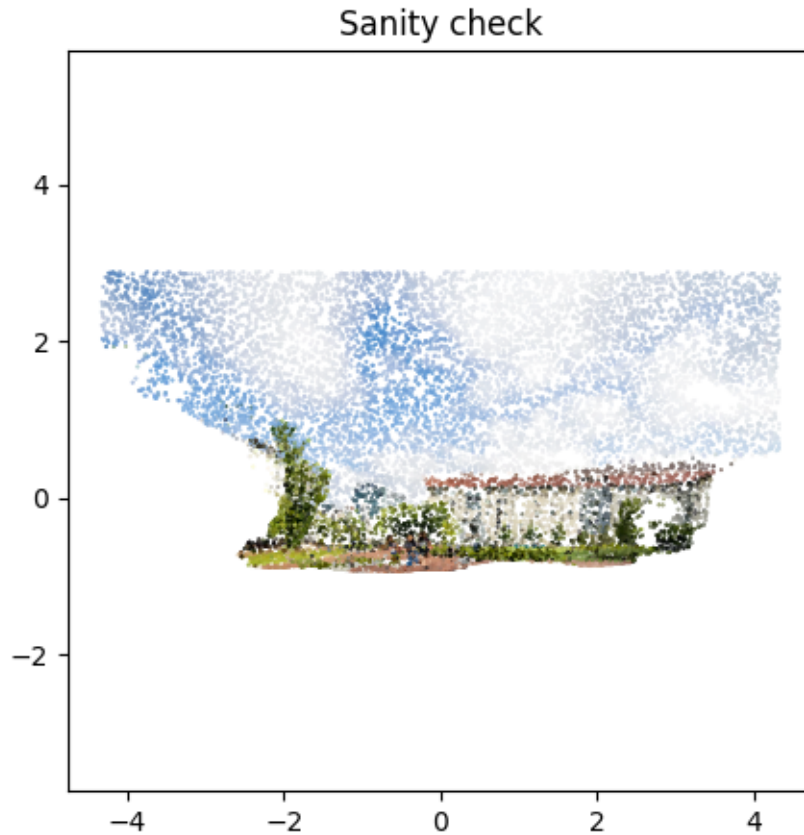
u, v = np.meshgrid(np.arange(Wd, dtype=np.float32),
                    np.arange(Hd, dtype=np.float32))

### YOUR CODE HERE
Z = 1.0 / (A + B * y)
X = (u - cx) * Z / fx
Y = (v - cy) * Z / fy
### END YOUR CODE

pts_cam = np.stack([X[valid], Y[valid], Z[valid]], axis=1)

K =
[[433.01270189  0.          750.          ]
 [ 0.          433.01270189 500.          ]
 [ 0.           0.           1.          ]]

[14]: # Quick 2D sanity check
flat_cols = rgb.reshape(-1,3)[valid.reshape(-1)]
idx = np.random.choice(pts_cam.shape[0], min(20000, pts_cam.shape[0]),
    ↪replace=False)
plt.scatter(pts_cam[idx,0], -pts_cam[idx,1], s=0.2, c=flat_cols[idx]/255.0)
plt.title("Sanity check"); plt.axis('equal'); plt.show()
```



Extract per-point color

```
[15]: cols = rgb.reshape(-1,3)[valid.reshape(-1)]
      print("Colors:", cols.shape)
```

Colors: (1462743, 3)

Save point cloud to PLY

```
[23]: def write_ply(path:str, xyz:np.ndarray, rgb:np.ndarray):

      """Write ASCII PLY: x y z r g b."""
      N = xyz.shape[0]
      if rgb.dtype != np.uint8:
          rgb = np.clip(rgb, 0, 255).astype(np.uint8)

      header = "\n".join([
          "ply", "format ascii 1.0",
          f"element vertex {N}",
          "property float x", "property float y", "property float z",
          "property uchar red", "property uchar green", "property uchar blue",
```



```

        "end_header"
    ]) + "\n"

    with open(path, "w") as f:
        f.write(header)
        for (x,y,z),(r,g,b) in zip(xyz, rgb):
            f.write(f"{x:.6f} {y:.6f} {z:.6f} {int(r)} {int(g)} {int(b)}\n")
    print("Saved:", os.path.abspath(path))

```

```

[24]: out_ply = "emory_campus_pcd.ply"
      write_ply(out_ply, pts_cam, cols)

      # Auto-download
      try:
          from google.colab import files
          files.download(out_ply)
      except:
          print("what")
          pass

```

Saved: /content/emory_campus_pcd.ply

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

1.4 Part D. Visualize the Point Cloud (10 points)

1.4.1 Explanation

- 1) Download **MeshLab** (lightweight, open-source) from <https://www.meshlab.net/>.
- 2) Open `emory_campus_pcd.ply`.
- 3) Adjust lighting settings (Tools → Options).
- 4) Hide the trackball (View → uncheck **Show Trackball**).
- 5) Take a screenshot by clicking the **camera** icon. Save it as `emory_campus_pcd_visualized.png`

Adjust to a bright lighting setting as shown in the image.

```

[25]: img_inst = cv2.imread('meshlab-adjust-lighting.png')
      img_inst = cv2.cvtColor(img_inst, cv2.COLOR_BGR2RGB)
      plt.imshow(img_inst)
      plt.axis('off')
      plt.show()

```

```

-----
error                                Traceback (most recent call last)
/tmp/ipython-input-1262014625.py in <cell line: 0>()

```

```

1 img_inst = cv2.imread('meshlab-adjust-lighting.png')
----> 2 img_inst = cv2.cvtColor(img_inst, cv2.COLOR_BGR2RGB)
3 plt.imshow(img_inst)
4 plt.axis('off')
5 plt.show()

error: OpenCV(4.12.0) /io/opencv/modules/imgproc/src/color.cpp:199: error: (-21 :
↳Assertion failed) !_src.empty() in function 'cvtColor'

```

Take a screenshot.

```

[26]: pcd_img = cv2.imread('emory_campus_pcd_visualized.png')
pcd_img = cv2.cvtColor(pcd_img, cv2.COLOR_BGR2RGB)
plt.imshow(pcd_img)
plt.axis('off')
plt.show()

```



1.5 Part E. Create a 3D Photo of Your Favorite Scene (10 points)

1.5.1 Explanation

- 1) Open your favorite outdoor scene photo.
- 2) Predict the relative (inverse) depth map using [Depth Anything V2](#). Export the **16-bit raw** depth image (e.g., `favorite_scene_depth_raw.png`).
- 3) Build a point cloud from your photo using the predicted depth and save it as `favorite_scene_pcd.ply`.
- 4) Visualize the point cloud and take a screenshot saved as `favorite_scene_pcd_visualized.png`.

```

[27]: # Input file path
rgb_path = "car.png"
depth_raw = "favorite_scene_depth_raw.png"

# --- Load RGB input ---
rgb_bgr = cv2.imread(rgb_path, cv2.IMREAD_COLOR)
assert rgb_bgr is not None, "Could not read RGB"
Hc, Wc = rgb_bgr.shape[:2]
show(rgb_bgr, "RGB")

# --- Load depth as relative (0..1) ---
depth_rel = None

r16 = cv2.imread(depth_raw, cv2.IMREAD_UNCHANGED)
if r16 is not None:
    if r16.dtype != np.uint16:
        raise ValueError("16-bit PNG must be uint16.")
    nonzero = r16[r16>0]
    denom = float(nonzero.max()) if nonzero.size else 1.0
    depth_rel = r16.astype(np.float32)/max(denom,1.0)
    show((np.clip(depth_rel,0,1)*255).astype(np.uint8), "Relative depth",
    ↪gray=True)

Hd, Wd = depth_rel.shape[:2]

# Align RGB to depth resolution
if (Hd,Wd)!=(Hc,Wc):
    rgb_bgr = cv2.resize(rgb_bgr, (Wd,Hd), interpolation=cv2.INTER_LINEAR)
    Hc, Wc = rgb_bgr.shape[:2]

```

RGB



Relative depth



```
[28]: def normalize_depth(x):
      x = x.astype(np.float32)
      mn, mx = float(x.min()), float(x.max())
      return (x - mn) / (mx - mn + 1e-8)

[29]: # --- Depth Anything v2 calibration: TrueDepth = 1 / (A + B * y), y in [0,1]
      ↪---
      y = normalize_depth(depth_rel) # norm_prediction

      # Pick two percentiles and assign plausible metric depths (m)
      p_lo, p_hi = 5.0, 95.0 # far-ish, near-ish
      yL, yH = np.percentile(y, [p_lo, p_hi])
      ZL_target = 2.5 # meters (far background)
      ZH_target = 0.8 # meters (near foreground)

      invZL, invZH = 1.0/ZL_target, 1.0/ZH_target

      ### YOUR CODE HERE
      B = (invZH - invZL) / (yH - yL)
      A = invZH - B * yH
```

```
depth_m = 1.0 / (A + B * y)
### END YOUR CODE
```

```
[30]: # Cleanup
valid = np.isfinite(depth_m) & (depth_m > 1e-6)
if valid.any():
    lo, hi = np.percentile(depth_m[valid], [1.0, 99.0])
    valid &= (depth_m >= lo) & (depth_m <= hi)

# Remove near-white background
rgb = cv2.cvtColor(rgb_bgr, cv2.COLOR_BGR2RGB)
valid &= (np.mean(rgb, axis=2) < 245)

print(f"Depth A={A:.6f}, B={B:.6f}")
print("Valid pixels:", int(valid.sum()), "/", valid.size)

# Range-normalized visualization
rng = np.ptp(depth_m[valid]) if valid.any() else 1.0
viz = (np.clip((depth_m - depth_m[valid].min()), 0, rng + 1e-8) / (rng + 1e-8), 0, 1) * 255).
      .astype(np.uint8)
show(viz, "Metric depth", gray=True)
```

Depth A=0.400000, B=0.967506
Valid pixels: 4201054 / 5529600

Metric depth



```

[31]: FOV_DEG = 120.0
K = build_intrinsics(Wd, Hd, fov_deg=FOV_DEG)
fx, fy, cx, cy = K[0,0], K[1,1], K[0,2], K[1,2]
print("K =\n", K)

u, v = np.meshgrid(np.arange(Wd, dtype=np.float32),
                    np.arange(Hd, dtype=np.float32))

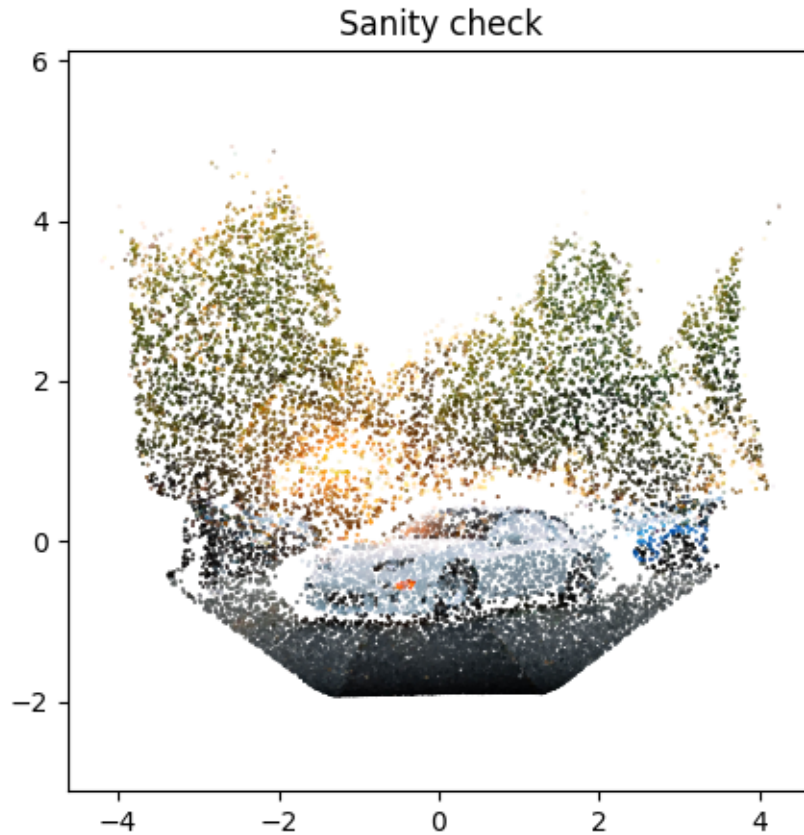
### YOUR CODE HERE
Z = 1.0 / (A + B * y)
X = (u - cx) * Z / fx
Y = (v - cy) * Z / fy
### END YOUR CODE

pts_cam = np.stack([X[valid], Y[valid], Z[valid]], axis=1)

K =
[[5.54256258e+02 0.00000000e+00 9.60000000e+02]
 [0.00000000e+00 5.54256258e+02 1.44000000e+03]
 [0.00000000e+00 0.00000000e+00 1.00000000e+00]]

[32]: # Quick 2D sanity check
flat_cols = rgb.reshape(-1,3)[valid.reshape(-1)]
idx = np.random.choice(pts_cam.shape[0], min(20000, pts_cam.shape[0]),
                        replace=False)
plt.scatter(pts_cam[idx,0], -pts_cam[idx,1], s=0.2, c=flat_cols[idx]/255.0)
plt.title("Sanity check"); plt.axis('equal'); plt.show()

```



```
[33]: cols = rgb.reshape(-1,3)[valid.reshape(-1)]
      print("Colors:", cols.shape)
```

Colors: (4201054, 3)

Save point cloud to PLY

```
[34]: def write_ply(path:str, xyz:np.ndarray, rgb:np.ndarray):

      """Write ASCII PLY: x y z r g b."""
      N = xyz.shape[0]
      if rgb.dtype != np.uint8:
          rgb = np.clip(rgb, 0, 255).astype(np.uint8)

      header = "\n".join([
          "ply", "format ascii 1.0",
          f"element vertex {N}",
          "property float x", "property float y", "property float z",
          "property uchar red", "property uchar green", "property uchar blue",
          "end_header"
      ]) + "\n"
```



```

with open(path, "w") as f:
    f.write(header)
    for (x,y,z),(r,g,b) in zip(xyz, rgb):
        f.write(f"{x:.6f} {y:.6f} {z:.6f} {int(r)} {int(g)} {int(b)}\n")
print("Saved:", os.path.abspath(path))

```

```

[35]: out_ply = "car.ply"
write_ply(out_ply, pts_cam, cols)

# Auto-download
try:
    from google.colab import files
    files.download(out_ply)
except:
    print("what")
    pass

```

Saved: /content/car.ply

<IPython.core.display.Javascript object>

<IPython.core.display.Javascript object>

```

[36]: !apt-get update -qq
      !apt-get install -y texlive-xetex texlive-fonts-recommended_
      ↪texlive-plain-generic pandoc
      !apt-get install -y pandoc
      !pip install nbconvert

```

W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (sources.list entry misspelt?)

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
 fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
 libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
 libsynchronet2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
 lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
 ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
 teckit tex-common tex-gyre texlive-base texlive-binaries texlive-latex-base
 texlive-latex-extra texlive-latex-recommended texlive-pictures tipa

xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex
default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 53 newly installed, 0 to remove and 39 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17
[33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]
Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64 2.13.1-1 [1,221 kB]
Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all 2.004.5-6.1 [4,532 kB]
Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]
Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems all 3.3.5-2ubuntu1.2 [228 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1 [5,100 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7 kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all 0.1.1-2 [12.6 kB]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all 0.3.2-1ubuntu0.1 [24.9 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsyntax2 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64 2.5.11+ds1-1 [421 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64 0.13.72+dfsg.1-1.1 [27.0 kB]

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Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-encodings all
1:1.0.5-0ubuntu2 [578 kB]
Get:36 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:37 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/main amd64 tiutils amd64
1.41-4build2 [61.3 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:42 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 182 MB in 14s (13.3 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126675 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...

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Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.

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Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.

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Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../38-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../45-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../46-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../47-texlive-latex-recommended_2021.20220204-1_all.deb ...

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Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../48-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../49-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../50-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../51-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../52-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...

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Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic

```

link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running updmap-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

 This may take some time... done.

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

0 upgraded, 0 newly installed, 0 to remove and 39 not upgraded.

Requirement already satisfied: nbconvert in /usr/local/lib/python3.12/dist-packages (7.16.6)

Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from nbconvert) (4.13.5)

Requirement already satisfied: bleach!=5.0.0 in /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert) (6.2.0)

Requirement already satisfied: defusedxml in /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.7.1)

Requirement already satisfied: jinja2>=3.0 in /usr/local/lib/python3.12/dist-

packages (from nbconvert) (3.1.6)
 Requirement already satisfied: jupyter-core>=4.7 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (5.8.1)
 Requirement already satisfied: jupyterlab-pygments in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.3.0)
 Requirement already satisfied: markupsafe>=2.0 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.0.3)
 Requirement already satisfied: mistune<4,>=2.0.3 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (3.1.4)
 Requirement already satisfied: nbclient>=0.5.0 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (0.10.2)
 Requirement already satisfied: nbformat>=5.7 in /usr/local/lib/python3.12/dist-
 packages (from nbconvert) (5.10.4)
 Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
 packages (from nbconvert) (25.0)
 Requirement already satisfied: pandocfilters>=1.4.1 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (1.5.1)
 Requirement already satisfied: pygments>=2.4.1 in
 /usr/local/lib/python3.12/dist-packages (from nbconvert) (2.19.2)
 Requirement already satisfied: traitlets>=5.1 in /usr/local/lib/python3.12/dist-
 packages (from nbconvert) (5.7.1)
 Requirement already satisfied: webencodings in /usr/local/lib/python3.12/dist-
 packages (from bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert) (0.5.1)
 Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
 /usr/local/lib/python3.12/dist-packages (from bleach[css]!=5.0.0->nbconvert)
 (1.4.0)
 Requirement already satisfied: platformdirs>=2.5 in
 /usr/local/lib/python3.12/dist-packages (from jupyter-core>=4.7->nbconvert)
 (4.5.0)
 Requirement already satisfied: jupyter-client>=6.1.12 in
 /usr/local/lib/python3.12/dist-packages (from nbclient>=0.5.0->nbconvert)
 (7.4.9)
 Requirement already satisfied: fastjsonschema>=2.15 in
 /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (2.21.2)
 Requirement already satisfied: jsonschema>=2.6 in
 /usr/local/lib/python3.12/dist-packages (from nbformat>=5.7->nbconvert) (4.25.1)
 Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-
 packages (from beautifulsoup4->nbconvert) (2.8)
 Requirement already satisfied: typing-extensions>=4.0.0 in
 /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->nbconvert)
 (4.15.0)
 Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-
 packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (25.4.0)
 Requirement already satisfied: jsonschema-specifications>=2023.03.6 in
 /usr/local/lib/python3.12/dist-packages (from
 jsonschema>=2.6->nbformat>=5.7->nbconvert) (2025.9.1)
 Requirement already satisfied: referencing>=0.28.4 in
 /usr/local/lib/python3.12/dist-packages (from

```

jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.36.2)
Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.12/dist-
packages (from jsonschema>=2.6->nbformat>=5.7->nbconvert) (0.27.1)
Requirement already satisfied: entrypoints in /usr/local/lib/python3.12/dist-
packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (0.4)
Requirement already satisfied: nest-asyncio>=1.5.4 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.6.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
/usr/local/lib/python3.12/dist-packages (from jupyter-
client>=6.1.12->nbclient>=0.5.0->nbconvert) (2.9.0.post0)
Requirement already satisfied: pyzmq>=23.0 in /usr/local/lib/python3.12/dist-
packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (26.2.1)
Requirement already satisfied: tornado>=6.2 in /usr/local/lib/python3.12/dist-
packages (from jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert) (6.4.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.8.2->jupyter-
client>=6.1.12->nbclient>=0.5.0->nbconvert) (1.17.0)

```

```
[37]: !jupyter nbconvert --to pdf "HW3.ipynb"
```

```

[NbConvertApp] WARNING | pattern 'HW3.ipynb' matched no files
This application is used to convert notebook files (*.ipynb)
to various other formats.

```

WARNING: THE COMMANDLINE INTERFACE MAY CHANGE IN FUTURE RELEASES.

Options

=====

The options below are convenience aliases to configurable class-options, as listed in the "Equivalent to" description-line of the aliases.

To see all configurable class-options for some <cmd>, use:

```
<cmd> --help-all
```

--debug

set log level to logging.DEBUG (maximize logging output)

Equivalent to: [--Application.log_level=10]

--show-config

Show the application's configuration (human-readable format)

Equivalent to: [--Application.show_config=True]

--show-config-json

Show the application's configuration (json format)

Equivalent to: [--Application.show_config_json=True]

--generate-config

generate default config file

Equivalent to: [--JupyterApp.generate_config=True]

-y

Answer yes to any questions instead of prompting.

Equivalent to: [--JupyterApp.answer_yes=True]

--execute

Execute the notebook prior to export.

Equivalent to: [--ExecutePreprocessor.enabled=True]

--allow-errors

Continue notebook execution even if one of the cells throws an error and include the error message in the cell output (the default behaviour is to abort conversion). This flag is only relevant if '--execute' was specified, too.

Equivalent to: [--ExecutePreprocessor.allow_errors=True]

--stdin

read a single notebook file from stdin. Write the resulting notebook with default basename 'notebook.*'

Equivalent to: [--NbConvertApp.from_stdin=True]

--stdout

Write notebook output to stdout instead of files.

Equivalent to: [--NbConvertApp.writer_class=StdoutWriter]

--inplace

Run nbconvert in place, overwriting the existing notebook (only relevant when converting to notebook format)

Equivalent to: [--NbConvertApp.use_output_suffix=False]

--NbConvertApp.export_format=notebook --FilesWriter.build_directory=]

--clear-output

Clear output of current file and save in place, overwriting the existing notebook.

Equivalent to: [--NbConvertApp.use_output_suffix=False]

--NbConvertApp.export_format=notebook --FilesWriter.build_directory=

--ClearOutputPreprocessor.enabled=True]

--coalesce-streams

Coalesce consecutive stdout and stderr outputs into one stream (within each cell).

Equivalent to: [--NbConvertApp.use_output_suffix=False]

--NbConvertApp.export_format=notebook --FilesWriter.build_directory=

--CoalesceStreamsPreprocessor.enabled=True]

--no-prompt

Exclude input and output prompts from converted document.

Equivalent to: [--TemplateExporter.exclude_input_prompt=True]

--TemplateExporter.exclude_output_prompt=True]

--no-input

Exclude input cells and output prompts from converted document.

This mode is ideal for generating code-free reports.

Equivalent to: [--TemplateExporter.exclude_output_prompt=True]

--TemplateExporter.exclude_input=True

--TemplateExporter.exclude_input_prompt=True]

--allow-chromium-download

Whether to allow downloading chromium if no suitable version is found on the system.

Equivalent to: [--WebPDFExporter.allow_chromium_download=True]

--disable-chromium-sandbox

Disable chromium security sandbox when converting to PDF..
Equivalent to: [--WebPDFExporter.disable_sandbox=True]

--show-input
Shows code input. This flag is only useful for dejavu users.
Equivalent to: [--TemplateExporter.exclude_input=False]

--embed-images
Embed the images as base64 dataurls in the output. This flag is only useful for the HTML/WebPDF/Slides exports.
Equivalent to: [--HTMLExporter.embed_images=True]

--sanitize-html
Whether the HTML in Markdown cells and cell outputs should be sanitized..
Equivalent to: [--HTMLExporter.sanitize_html=True]

--log-level=<Enum>
Set the log level by value or name.
Choices: any of [0, 10, 20, 30, 40, 50, 'DEBUG', 'INFO', 'WARN', 'ERROR', 'CRITICAL']
Default: 30
Equivalent to: [--Application.log_level]

--config=<Unicode>
Full path of a config file.
Default: ''
Equivalent to: [--JupyterApp.config_file]

--to=<Unicode>
The export format to be used, either one of the built-in formats
['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf']
or a dotted object name that represents the import path for an
``Exporter`` class
Default: ''
Equivalent to: [--NbConvertApp.export_format]

--template=<Unicode>
Name of the template to use
Default: ''
Equivalent to: [--TemplateExporter.template_name]

--template-file=<Unicode>
Name of the template file to use
Default: None
Equivalent to: [--TemplateExporter.template_file]

--theme=<Unicode>
Template specific theme(e.g. the name of a JupyterLab CSS theme distributed as prebuilt extension for the lab template)
Default: 'light'
Equivalent to: [--HTMLExporter.theme]

--sanitize_html=<Bool>
Whether the HTML in Markdown cells and cell outputs should be sanitized. This should be set to True by nbviewer or similar tools.
Default: False
Equivalent to: [--HTMLExporter.sanitize_html]

```

--writer=<DottedObjectName>
    Writer class used to write the
                                results of the conversion
    Default: 'FilesWriter'
    Equivalent to: [--NbConvertApp.writer_class]
--post=<DottedOrNone>
    PostProcessor class used to write the
                                results of the conversion
    Default: ''
    Equivalent to: [--NbConvertApp.postprocessor_class]
--output=<Unicode>
    Overwrite base name use for output files.
                                Supports pattern replacements '{notebook_name}'.
    Default: '{notebook_name}'
    Equivalent to: [--NbConvertApp.output_base]
--output-dir=<Unicode>
    Directory to write output(s) to. Defaults
                                to output to the directory of each notebook.
To recover
                                previous default behaviour (outputting to the
current
                                working directory) use . as the flag value.
    Default: ''
    Equivalent to: [--FilesWriter.build_directory]
--reveal-prefix=<Unicode>
    The URL prefix for reveal.js (version 3.x).
    This defaults to the reveal CDN, but can be any url pointing to a
copy
    of reveal.js.
    For speaker notes to work, this must be a relative path to a local
    copy of reveal.js: e.g., "reveal.js".
    If a relative path is given, it must be a subdirectory of the
    current directory (from which the server is run).
    See the usage documentation
    (https://nbconvert.readthedocs.io/en/latest/usage.html#reveal-js-
html-slideshow)
    for more details.
    Default: ''
    Equivalent to: [--SlidesExporter.reveal_url_prefix]
--nbformat=<Enum>
    The nbformat version to write.
    Use this to downgrade notebooks.
    Choices: any of [1, 2, 3, 4]
    Default: 4
    Equivalent to: [--NotebookExporter.nbformat_version]

```

Examples

The simplest way to use nbconvert is

```
> jupyter nbconvert mynotebook.ipynb --to html
```

Options include ['asciidoc', 'custom', 'html', 'latex', 'markdown', 'notebook', 'pdf', 'python', 'qtpdf', 'qtpng', 'rst', 'script', 'slides', 'webpdf'].

```
> jupyter nbconvert --to latex mynotebook.ipynb
```

Both HTML and LaTeX support multiple output templates. LaTeX includes

'base', 'article' and 'report'. HTML includes 'basic', 'lab' and 'classic'. You can specify the flavor of the format used.

```
> jupyter nbconvert --to html --template lab mynotebook.ipynb
```

You can also pipe the output to stdout, rather than a file

```
> jupyter nbconvert mynotebook.ipynb --stdout
```

PDF is generated via latex

```
> jupyter nbconvert mynotebook.ipynb --to pdf
```

You can get (and serve) a Reveal.js-powered slideshow

```
> jupyter nbconvert myslides.ipynb --to slides --post serve
```

Multiple notebooks can be given at the command line in a couple of different ways:

```
> jupyter nbconvert notebook*.ipynb
> jupyter nbconvert notebook1.ipynb notebook2.ipynb
```

or you can specify the notebooks list in a config file, containing::

```
c.NbConvertApp.notebooks = ["my_notebook.ipynb"]
```

```
> jupyter nbconvert --config mycfg.py
```

To see all available configurables, use `--help-all`.

[]: